

preamble_test

2026-09-08

Exercises and objectives

The objectives of today's exercises are:

1) Check that your environment works by running single level ... 2) ... and multilevel models in R 3) Run a single level model in Python

We are going to fit some single level and multilevel models on the *sleepstudy* data from the *lme4* package. Check out the data description and format using `?sleepstudy`

```
library(lme4)
data("sleepstudy")
```

Exercise 1

Do a linear regression, modelling reaction time as dependent of number of days of sleep deprivation using the function *lm*

(The General Linear Model: $Y = X\beta + \epsilon; \epsilon \sim N(0, \sigma^2)$)

```
single.level.model <- lm(formula=..., data=...)
```

1. Extract $\hat{\beta}$, Y , $\hat{Y}$, x , X , $\hat{\sigma}$ and ϵ from *single.level.model* (hint: have a look at the function *model.matrix*)
 - i. create a plot that illustrates x , Y , $\hat{Y}$ and ϵ for subject 308)
2. Estimate β for a quadratic model ($y = \beta_2 x^2 + \beta_1 x + \beta_0$) *without* using *lm*; $\hat{\beta} = (X^T X)^{-1} X^T Y$ (hint: add a third column to X from step 1)
 - i. compare your acquired $\hat{\beta}$ with the output of a corresponding quadratic model created using *lm* (hint: use the function *I*, see details under help and the sub-section formula operators here: <https://www.datacamp.com/community/tutorials/r-formula-tutorial>)
3. Plot the data points using `plot(Reaction ~ Days, data=sleepstudy)`, then add the fitted lines from your linear model and from your quadratic model
 - i. Calculate the sum of squares for each of the two models, where n is the number of observations

$$SS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- ii. Calculate the residual standard deviation, $\hat{\sigma} = \sqrt{\frac{SS}{n-k}}$ where n is the number of observations and k is the number of parameters fitted, for the two models (remember this is the estimated *sigma* in $\epsilon \sim N(0, \sigma^2)$)
 - iii. Based on the plot, the sums of squares and the $\hat{\sigma}$'s, which of the two models, do you prefer?
4. Plot the design matrix, X from the quadratic model, using *image.plot* from the *fields* package using the following code and explain what you see:

```
library(fields)
image.plot(x=1:dim(X)[2], 1:dim(X)[1], t(X), xlab='Predictors',
```

```

        ylab='Observations', xaxt='n', main='Design matrix: X')
axis(1, 1:dim(X)[2], 1:dim(X)[2])

```

Exercise 2

Do a multilevel linear regression, modelling reaction time as dependent of number of days of sleep deprivation, while accounting for the subject structure by modelling subject-level intercepts and slopes using the function *lmer*

(Multilevel modelling: $Y = X\beta + Zb + \epsilon$; $\epsilon \sim N(0, \sigma^2)$; $b \sim N(0, \Sigma)$; Σ is a covariance matrix)

```

library(lme4)
multilevel.model <- lmer(formula=..., data=...)

```

1. Extract $\hat{\beta}$, Y , $\hat{Y}$, X , Z , $\hat{b}$, $\hat{\Sigma}$, $\hat{\sigma}$ and ϵ from *multilevel.model* (hint: use the function *getME*)
 - i. create a plot that illustrates the fitted lines for each subject
2. Plot the design matrices, X and Z using *image.plot* (note: you will have to apply *as.matrix* to Z as *image.plot* doesn't handle sparse matrices)
 - i. Are there any differences between the current X and the X from Exercise 1.1 - why or why not?
 - ii. Explain in your own words what Z illustrates
3. Finally, calculate the sum of squares and compare with the sum of squares for the *single.level.model*

Exercise 3

Check whether your *Python* works within your favourite notebook IDE, *.ipynb*

Access the California Housing dataset

```

from sklearn.datasets import fetch_california_housing

california = fetch_california_housing()
x = california.data # independent variables; check california.feature_names
y = california.target # dependent variable: price actually sold for

```

1. Run a single-level linear regression on the California Housing dataset from *scikit-learn* using algebra;
 - i. Build a model matrix X , with 2 predictors, an average (intercept) and a slope, predicting the prices houses were sold for based on the median income (MedInc) of the block (first column of *california_data*)
2. Plot the fitted regression line and Y and using *plt.plot*

Chunks for inspiration

```

import numpy as np
X = np.zeros(shape=(n_obs, n_predictors)) ## what are the n's?
X[:, 0] = ?# what should the entries in the first column be?
X[:, 1] = ?# what should the entries in the second column be?
beta_hat = ?# @ is matrix multiplication and np.linalg.inv is for inverting a matrix

```

```

import matplotlib.pyplot as plt
plt.figure() # create figure
plt.plot(x[:, 0], y, 'o') # add points
x_pred = np.array([0, np.max(x[:, 0])])
y_pred = ?
plt.plot(x_pred, y_pred) ## add a line
plt.xlabel() ## add meaningful labels and a title
plt.ylabel()
plt.title()
plt.show() ## actually showing the figure

```